

Summary of automated booklet-printing pipeline

Overview

This automation removes the manual effort of formatting booklets. Every time you push a standard document to GitHub, a cloud script instantly rearranges it into a side-by-side, printable landscape layout. Your home computer then silently downloads this file the moment it boots up and connects to the internet.

The Automation Workflow

Step 1 (Upload): You save your document as a PDF and push it to your GitHub repository.

Step 2 (Convert): GitHub automatically triggers a Python cloud worker that creates a fresh landscape layout.

Step 3 (Merge): The cloud script glues page 1 to the left and page 2 to the right, saving a new printable file.

Step 4 (Sync): When you turn on your PC, a background script waits for Wi-Fi and pings GitHub for the file.

Step 5 (Check): The script compares the files; if you already have the latest version, it skips the download.

Step 6 (Pull): If a newer document exists, it pulls the file straight to your local folder, ready for you to print and fold.

Automation workflow broken down into chronological steps:

Phase 1: Creating PDF

Write Document: Edit bulletin in GDocs.

Export PDF: Using GoogleAppsScript to convert and commit and push to GitHub

Phase 2: The GitHub Cloud Automation

Detect Push: GitHub Actions notices a new PDF has been uploaded into the repository.

Boot Environment: GitHub automatically launches a secure virtual Linux container.

Install Python: The workflow environment quickly installs Python and the pypdf library.

Rearrange Document: The python script takes your original PDF pages and builds a blank landscape canvas.

Merge Side-by-Side: The script pastes page 1 on the left side and page 2 on the right side.

Commit Back: GitHub pushes this newly generated printable_booklet.pdf back into your repo files.

Phase 3: The Local Computer Startup Sync

Power On: You turn on your home computer or laptop connected to your printer.

Launch Script: The computer logs in and automatically launches the python background sync script.

Ping Network: The script silently runs a loop, waiting until your computer successfully connects to Wi-Fi/Ethernet.

Check Repository: Once connected, the script contacts GitHub and looks up the file size of the printable PDF.

Evaluate Changes: The script compares the online file size against your local folder copy.

Neglect/Skip: If the size is exactly the same, the script assumes you already have it and shuts down.

Pull Update: If the file is missing or has a new file size, it pulls the new booklet straight to your local directory and opens the directory.

Print: Print the updated file, and fold it in half.